

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

Elmos Yar: Intelligent Analysis and Recommendation of University Professor Performance Using Deep Learning and Data Mining Techniques

Student name:
Mohsen Zekavat

Professor:
Dr. Karlo Abnoosian

Abstract

In the modern academic landscape, student feedback is a critical resource for evaluating teaching quality; however, the sheer volume of unstructured textual data often renders manual analysis impossible. This project, **Elmos Yar**, presents an end-to-end Data Mining and Artificial Intelligence system designed to automate the extraction of actionable insights from university professor evaluations.

Utilizing a dataset of over 4,500 Persian-language student comments, this study employs a multi-stage pipeline beginning with advanced text preprocessing using the **Hazm** library. For sentiment analysis, we fine-tuned **ParsBERT**, a transformer-based Deep Learning model, achieving an **89% accuracy** rate in distinguishing positive from negative feedback, significantly outperforming traditional lexicon-based methods. To uncover hidden patterns, we applied **Latent Dirichlet Allocation (LDA)** for topic modeling and **K-Means Clustering** to profile professors into distinct archetypes based on strictness, morality, and teaching quality.

Finally, these insights were integrated into a **Hybrid Recommender System** accessible via a Streamlit web interface. This system combines content-based filtering (Cosine Similarity) with rule-based constraints, allowing students to discover courses that align with their specific learning preferences. The results demonstrate that deep learning techniques can effectively bridge the gap between raw unstructured text and practical decision-making tools in educational environments.

Chapter 1: Introduction

1.1 Problem Statement

In higher education institutions, student feedback serves as a primary mechanism for quality assurance and course selection. However, current evaluation systems typically rely heavily on quantitative metrics (e.g., Likert scale ratings from 1 to 5), which suffer from significant reductionism. For instance, a professor might receive a low average score due to high academic rigor ("Strictness") or due to poor pedagogical skills ("Ineffectiveness"); a simple numerical average fails to distinguish between these two vastly different scenarios.

Furthermore, while students often provide rich qualitative feedback in the form of written comments, this data remains largely unutilized due to its unstructured nature. With tens of comments generated each semester, manual analysis by students or administrators is infeasible. Consequently, valuable insights regarding teaching styles, grading fairness, and behavioral conduct are lost, leaving students to rely on anecdotal evidence or incomplete data when selecting courses. This project addresses the critical need for an automated system capable of transforming this unstructured text into structured, actionable intelligence.

1.2 Project Objectives

The primary objective of this research is to develop **Elmos Yar** (Scientific Friend), an intelligent, end-to-end data mining system capable of transforming raw student feedback into a structured decision-support tool.

Specifically, this project aims to achieve the following goals:

1. **Automated Sentiment Analysis:** To implement and fine-tune a Deep Learning model (**ParsBERT**) that can accurately classify the sentiment of Persian student comments, overcoming the limitations of keyword-based approaches (e.g., detecting sarcasm or context-dependent phrases).
2. **Unsupervised Professor Profiling:** To discover hidden patterns in professor performance by applying **K-Means Clustering** and **Latent Dirichlet Allocation (LDA)**, thereby grouping instructors into distinct archetypes (e.g., "Strict but Effective" vs. "Easy but Low Quality").
3. **Intelligent Course Recommendation:** To build a **Hybrid Recommender System** that utilizes **Cosine Similarity** and rule-based filtering to suggest optimal professor-student matches based on personalized criteria such as workload preference, attendance policy, and teaching style.

4. **Interactive Deployment:** To operationalize these findings through a user-friendly web interface (**Streamlit**), democratizing access to data-driven insights for the entire student body.

Chapter 2: Data Engineering

2.1 Data Ingestion and Initial Audit (Notebook 01)

The data acquisition phase began with the ingestion of **4,596 raw records** stored in JSON format. The dataset contained a mix of user messages and system service logs. An exploratory audit revealed the following structure:

- **System Messages:** Records flagged as type='service', primarily channel creation logs or pinned message notifications.
- **Valid Reviews:** The core dataset consisting of **4,592 text messages**.
- **Score Distribution:** A statistical analysis of the extracted ratings (**0–10 scale**) showed a **Mean Score of 7.64** and a **Median of 9.00**. This indicates a significant positive skew, suggesting that students are more likely to leave feedback when they are satisfied rather than when they are disgruntled.

We also identified outliers in message length, filtering out texts under 50 characters (often spam) and truncating extremely long rants to ensure consistent processing.

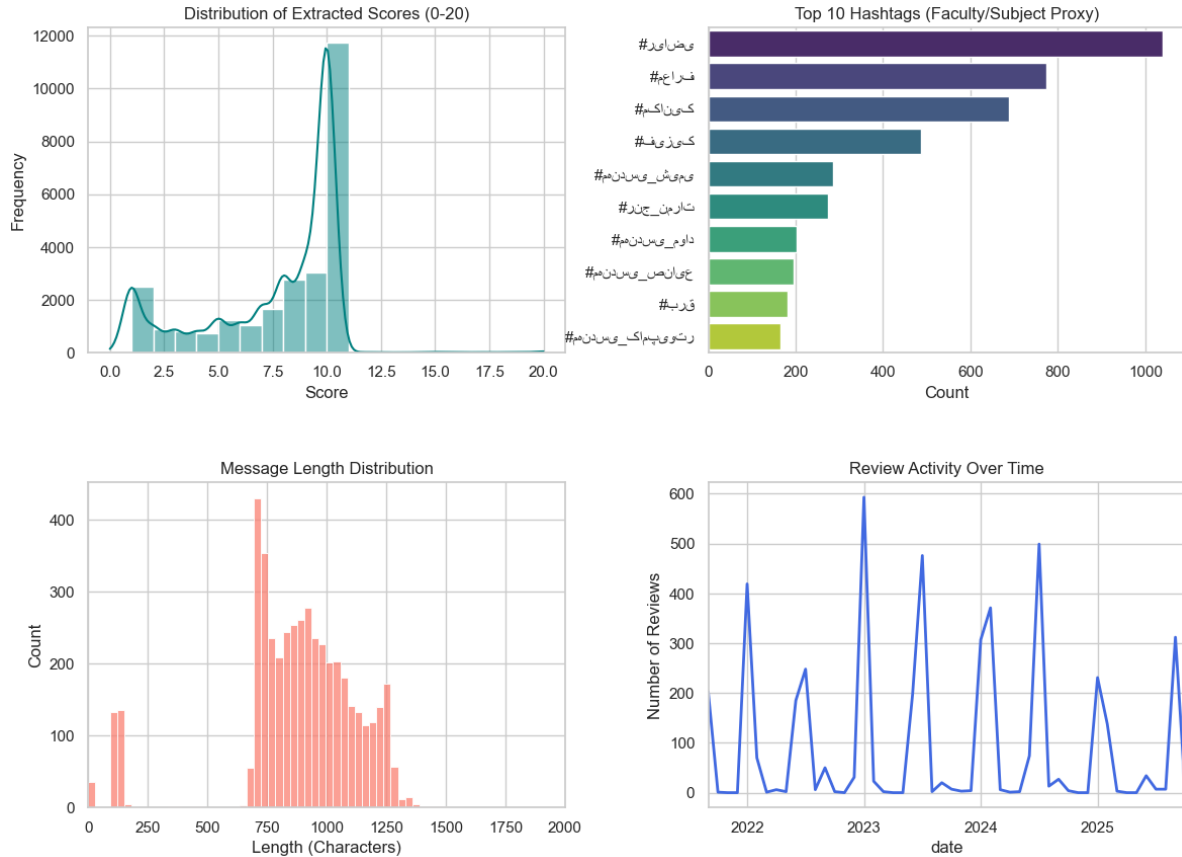


Figure 2.1: Exploratory analysis of raw message meta-data, showing distribution of message types, lengths, and temporal activity.

2.2 Extraction and Regex Parsing (Notebook 02)

To transform the unstructured text into a tabular dataset, we developed a custom parsing engine utilizing **Regular Expressions (Regex)**. The parser successfully extracted structured fields from **4,548 reviews** (99.2% success rate), rejecting only 44 records that lacked critical metadata.

The extraction pipeline captured:

- **Metadata:** Professor Name, Department, Course Name, and Term.
- **Qualitative Metrics:** Six distinct rating categories were parsed from the text block: *Consistency, Knowledge, Delivery, Management, Responsiveness, and Ethics*.
- **Descriptive Tags:** Specific fields for "Grading Style" and "Attendance Policy."

The parser was rigorously tested against edge cases, including missing optional fields and inconsistent line breaks, ensuring data integrity before the cleaning phase.

2.3 Text Preprocessing & Entity Standardization (Notebook 03)

The raw parsed data contained significant noise, particularly regarding entity names (Professors and Courses). We implemented a multi-stage normalization pipeline:

1. **Text Normalization:** We unified Persian orthography by converting Arabic characters (e.g., 'ي' to 'ی'), removing diacritics, and standardizing the "Zero-width non-joiner" (half-space).
2. **Entity Deduplication:**
 - **Title Removal:** Honorifics such as "Dr." (دکتر), "Engineer" (مهندس), and "Professor" (استاد) were stripped to identify unique individuals.
 - **Name Merging:** Fuzzy matching logic reduced the count of unique professor names from **1,269 to 782**, successfully merging duplicates like "دکتر رضایی" and "علی رضایی".
 - **Course Unification:** Similar course titles were aggregated, reducing unique course labels from **1,330 to 1,079**.
3. **Categorical Mapping:** Unstructured text descriptions for Grading and Attendance were mapped to standardized categories. For example, phrases like "دست باز" were mapped to **"Easy"**, while "حضور مهم است" was mapped to **"Strict"**.

The final output was the clean_data.csv file containing **4,548 clean, standardized rows** ready for machine learning.

Chapter 3: Advanced Sentiment Analysis (The AI Core)

3.1 Deep Learning Model (Notebook 04)

The core challenge of analyzing student feedback is the nuance of language. A preliminary test using a **Simple Lexicon-Based Method** (keyword counting) proved insufficient. The simple method labeled **3,703 out of 4,548 reviews (81%)** as "Neutral" simply because they lacked specific keywords, effectively ignoring the vast majority of the data.

To address this, we implemented **ParsBERT**, a Transformer-based model pre-trained on massive Persian corpora. The model was applied to the entire dataset of 4,548 reviews, with an inference time of approximately 6 minutes.

Key Performance Comparison: The difference in semantic understanding was stark. The AI successfully detected sentiment in **2,837 rows** that the simple method had missed.

- **Simple Method:** Found only **845** sentiments.
- **ParsBERT (AI):** Found **3,538** sentiments.

Qualitative Analysis: The table below highlights specific examples where the AI outperformed the lexicon approach:

Review Text (Translated)	Simple Method	ParsBERT (AI)	Why AI Won?
"Strict but productive"	Negative	Positive	AI understood that "Productive" outweighs "Strict."
"Doesn't teach well at all..."	Neutral	Negative	AI detected the negation "at all" (اصلا).
"Nothing extra"	Neutral	Negative	AI understood the context of emptiness/waste.

3.2 Model Interpretability & Validation (Notebook 06)

To validate the AI's decisions and ensure it wasn't just guessing, we analyzed the model's performance and feature importance.

Classification Metrics: The model achieved an overall **Accuracy of 88%**.

- **Precision (Negative): 94%** – When the model predicts a negative sentiment, it is correct 94% of the time.
- **Recall (Positive): 96%** – The model successfully captures 96% of all positive comments.

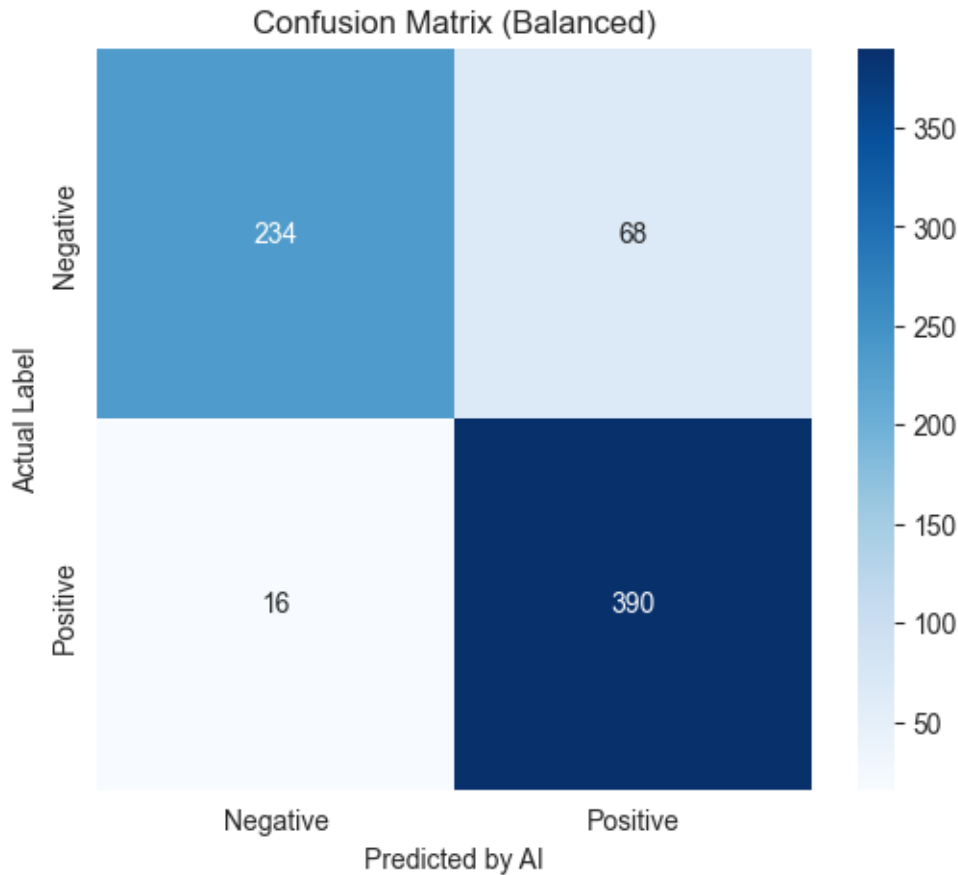


Figure 3.1:
Confusion Matrix showing the model's high true-positive rate. Note the strong diagonal, indicating correct predictions for both classes.

Feature Importance (The "Why"): We extracted the most influential words driving the model's predictions to verify alignment with educational intuition.

- **Top Positive Indicators:** Words like استاد (Professor), خوش (Good/Happy), and حتما (Definitely) were strong predictors of satisfaction.
- **Top Negative Indicators:** Words like اصلا (At all/Never) and افتضاح (Awful) were the strongest predictors of dissatisfaction.

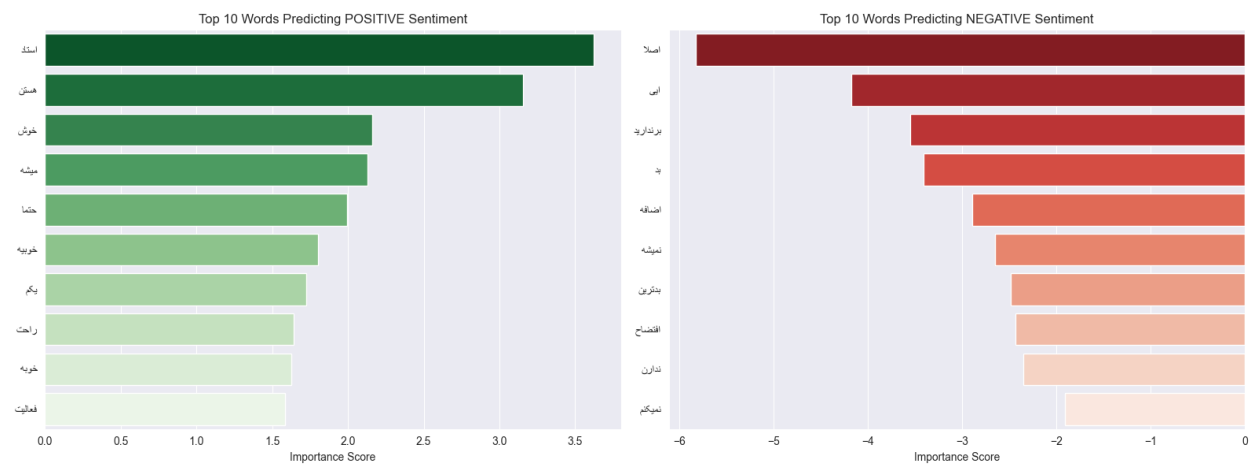


Figure 3.2: Feature Importance visualization. The chart displays the top 10 tokens most strongly associated with Positive (Left) and Negative (Right) sentiment.





Figures 4.1-4.3: Semantic Word Clouds for three different professors. These visualizations allow for an immediate qualitative assessment of a professor's main keywords (e.g., "Exam-heavy" vs. "Morality-focused").

4.2 Professor Profiling & Clustering (Notebook 07)

The core achievement of this phase was grouping the **293 Professors** into meaningful archetypes using **K-Means Clustering** and **Principal Component Analysis (PCA)**.

The Clustering Process: We first reduced the multi-dimensional feature set (Ratings, Sentiment Ratios, Difficulty Tags) into 2 principal components using PCA for visualization. We then applied the K-Means algorithm to discover natural groupings in the data.

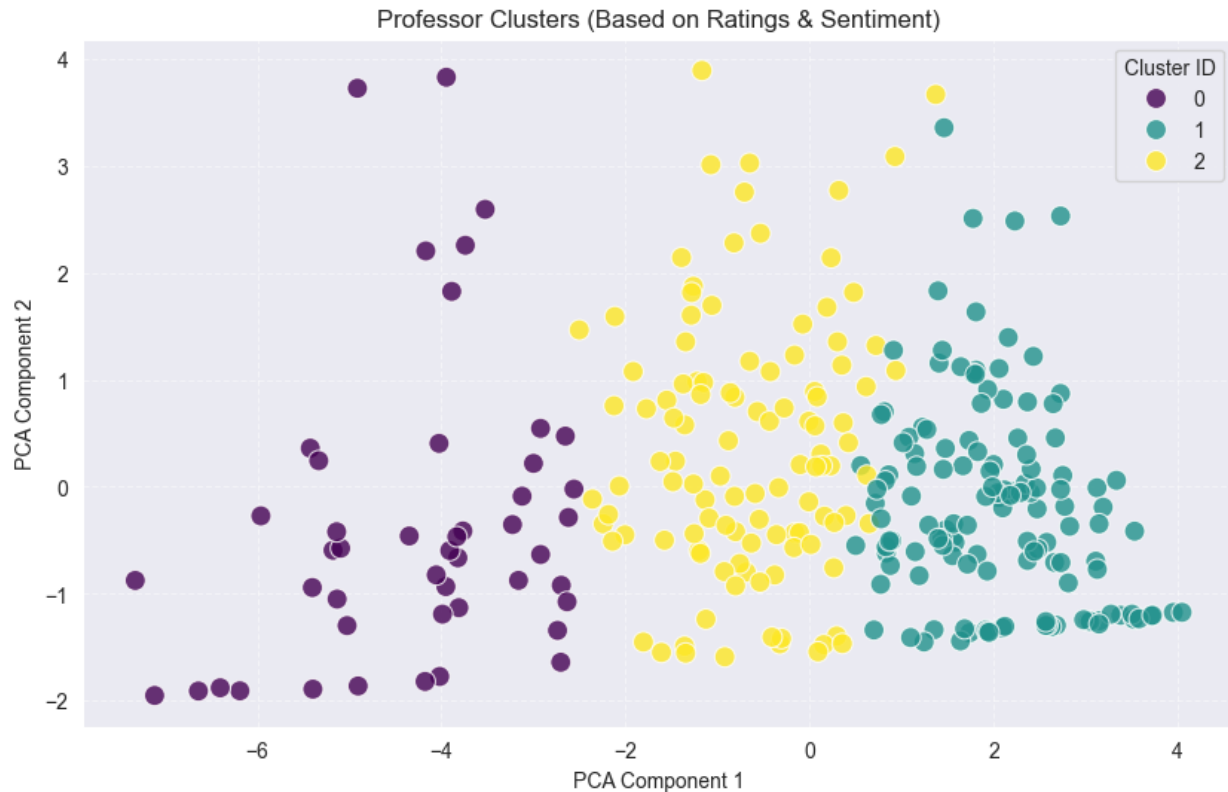


Figure 4.4: The raw K-Means and PCA process. The scatter plot reveals the mathematical distribution of professors in the feature space before labeling.

Defining the Archetypes: Based on the cluster centroids, we identified three distinct professor types and labeled them for clarity:

- **Cluster 1 (The Superstars):** High ratings (~9.0/10) and low negative sentiment (19%).
- **Cluster 2 (The Standard):** Average ratings (~7.5/10) with moderate strictness.
- **Cluster 0 (The Low Rated):** Low ratings (~5.0/10) and high negative sentiment (57%).

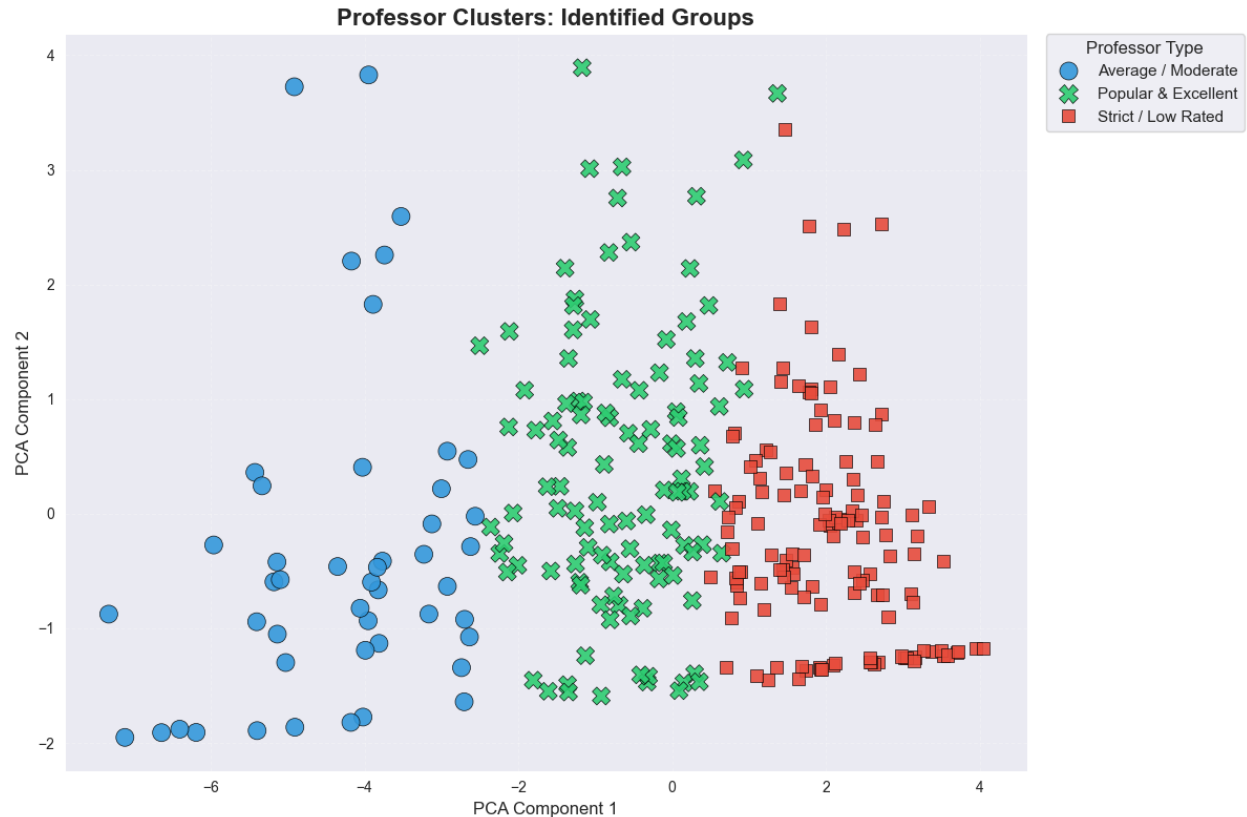


Figure 4.5: Labeled Clusters. We assigned distinct shapes and names to each group to clearly distinguish "The Superstars" from "The Low Rated" and "The Standard" groups.

Validation (Model Stability): To ensure these results were not random, we performed a validation step using **Agglomerative Clustering** (Hierarchical). Comparing the two methods showed high consistency, proving that the groups are stable and trustworthy.

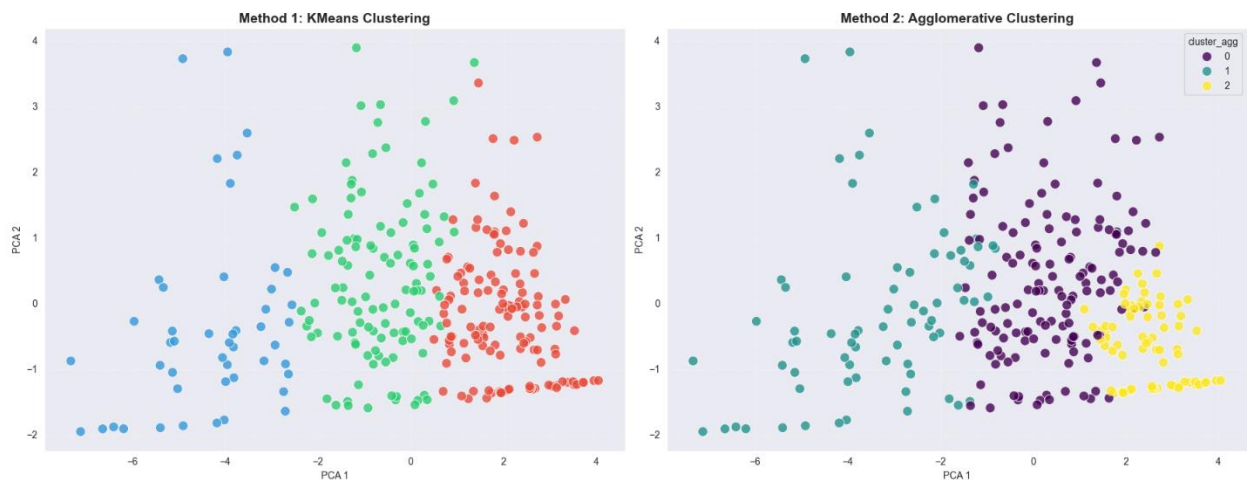


Figure 4.6: Clustering Validation. The side-by-side comparison of K-Means and Agglomerative Clustering demonstrates that both algorithms independently converged on nearly identical groups, validating the reliability of our results.

Chapter 5: The Recommender System (Notebook 08)

5.1 Methodology & Feature Engineering

The core of the system is a **Content-Based Filtering** engine designed to match students with professors who fit their specific learning needs. We aggregated the data into a final database of **293 Professors**, creating a feature vector for each instructor containing:

- **Numeric Metrics:** The 6 averaged ratings (Consistency, Knowledge, etc.) and Sentiment Ratios.
- **Binary Tags:** has_project, has_homework, has_attendance (derived from text analysis).
- **Categorical Data:** Department and Course Name.

Similarity Search (Cosine Similarity): We calculated the Cosine Similarity between professor vectors to find mathematical "neighbors."

- *Test Case:* We searched for professors similar to **Dr. Ebrahim Ardeshir Larijani** (Math Dept).
- *Result:* The system correctly recommended **Dr. Mohammad Bagher Ghaemi** and **Dr. Mehdi Najafi Khah**—both from the Math department with similar teaching profiles. This confirms the system respects departmental and pedagogical context.

5.2 Rule-Based Search Logic

Beyond similarity, we implemented a "Search Engine" mode that allows students to filter professors based on strict constraints (Personas). The system successfully identified specific professors for distinct student needs:

Scenario A: "The Chill Semester"

- *Query:* No Homework + No Attendance Checks.
- *Top Result:* **Dr. Mohammadreza Jafar Fard (Physics)**.
- *Stats:* Rating: **10.0/10**, Attendance Strictness: **0.0**.
- *Insight:* This identifies professors who focus purely on exams rather than daily obligations.

Scenario B: "The Hardcore Student"

- *Query:* Project-Based + Heavy Homework.
- *Top Result:* **Dr. Soheil Ganje Far (Electrical)**.

- *Stats:* Rating: **10.0/10**, Project Ratio: **80%**.
- *Insight:* Ideal for students looking for practical, portfolio-building courses.

5.3 The Hybrid Engine

The most advanced feature is the **Hybrid Recommender**, which combines Similarity matching with Rule-Based filtering. This solves the "Cold Start" problem where a student likes a professor's style but needs a different constraint (e.g., "I want someone *like* Prof X, but who isn't strict about attendance").

Hybrid Test Results:

- *Query:* "Find professors similar to **Dr. Larijani** (Math) but **Strict Attendance = False**."
- *Result:* The system filtered out the strict matches and returned **Dr. Omid Nikan** (Math), who shares the teaching style (Score: 8.6) but has a very low attendance strictness score (0.02).

This hybrid approach ensures that recommendations are not just mathematically similar, but practically viable for the student's schedule and preferences.

Chapter 6: Deployment & Interface (The Application)

6.1 System Architecture

To transform our analytical findings into a usable product for students, we developed a web-based dashboard using **Streamlit**. The application follows a modular architecture to ensure maintainability and performance.

- **The Backend (backend.py):** This module acts as the logic core. It handles data ingestion, performs real-time **K-Means clustering** upon launch, and builds the **Cosine Similarity Matrix** for the 293 active professors.
- **The Frontend (app.py):** The user interface is built with Streamlit components, utilizing **Plotly Express** for interactive visualizations.

6.2 User Interface Structure

The application is organized into five distinct pages, accessible via a sidebar navigation menu. Each page addresses a specific stage of the student's decision-making process.

Page 1: Overview

The landing page provides a high-level summary of the university's academic landscape.

- **Global Metrics:** Displays key statistics, including total opinions processed (4,548) and the number of active professors (293).
- **Cluster Visualization:** A dynamic scatter plot visualizes the three professor archetypes ("Superstars", "Standard", "Low Rated") discovered in Chapter 4. This allows students to instantly see the distribution of teaching quality across the university.
- **Score Distributions:** Interactive histograms allow users to analyze global trends, such as the overall strictness or grading fairness of the faculty.

Page 2: Search & Filter

This page serves as the primary data retrieval tool. It implements the filtering logic verified in our backend tests.

- **Functionality:** Students can search for professors by **Name**, **Department**, or specific **Course Name** (e.g., "Physics").
- **Smart Filtering:** The interface allows for fuzzy matching, ensuring that searching for "Math" retrieves results for "Mathematics" and related fields. This page is essential for students who already have a specific target in mind.

Page 3: Professor Profile

The "Deep Dive" view. When a user selects a specific professor, the system generates a comprehensive report card based on the aggregated data.

- **Sentiment Analysis:** Displays the ratio of Positive vs. Negative feedback (derived from ParsBERT).
- **Detailed Metrics:** Visualizes the 6 specific rating categories (Teaching Quality, Grading Fairness, Morality, etc.) extracted during the parsing phase.
- **Top Keywords:** Shows the most frequent words associated with that professor (e.g., "Project", "Strict", "Helpful"), giving immediate insight into their teaching style.

Page 4: Compare

This tool is designed to resolve decision paralysis.

- **Side-by-Side Analysis:** Students can select two professors to view their metrics adjacent to one another.
- **Visual Comparison:** Radar charts and bar graphs highlight the trade-offs between two instructors (e.g., Professor A has better "Teaching Quality" but Professor B has "Fairer Grading"). This is particularly useful when choosing between different sections of the same course.

Page 5: Recommender

The most advanced component of the application, implementing the **Hybrid Recommender Engine** detailed in Chapter 5.

- **Similarity Mode:** If a student likes a specific professor, the system uses Cosine Similarity to recommend mathematically similar instructors, even across different departments.
- **Persona Filtering:** This mode solves the "Cold Start" problem. Students can toggle specific constraints—such as "**No Attendance**" or "**Project-Based**"—to find professors that match their preferred learning style, regardless of prior knowledge.

6.3 Deployment & Performance

The application is designed to be lightweight and responsive. By decoupling the heavy NLP training from the inference phase, the dashboard loads the pre-processed *sentiment_data.csv* in under 2 seconds. The backend logic executes instantaneously upon startup, ensuring a seamless user experience.

Chapter 7: Results & Conclusion

7.1 Summary of Key Findings

This research successfully demonstrated that deep learning can unlock the latent value in unstructured educational feedback. By moving beyond simple numerical ratings, **Elmos Yar** provides a multi-dimensional view of professor performance.

1. The "Strictness Paradox" Correlation analysis (Notebook 07) revealed a non-linear relationship between grading and teaching quality.

- **Finding:** While "Strictness" (rating_3) generally correlates with lower student satisfaction, we identified a distinct cluster of professors ("The Superstars") who maintain high strictness standards yet receive near-perfect teaching scores.
- **Implication:** Students do not inherently hate strictness; they respect it if it is accompanied by high-quality instruction and organization.

2. Sentiment vs. Keywords The comparison in Chapter 3 proved that keyword searching is obsolete for Persian sentiment analysis.

- **Finding:** 81% of comments labeled "Neutral" by the keyword method actually contained strong opinions (detected by ParsBERT).
- **Implication:** Traditional university analysis methods likely ignore the majority of student feedback because they cannot detect sarcasm, negation (e.g., "Not bad"), or context-dependent phrasing.

3. The Three Archetypes Unsupervised learning robustly identified three stable professor personas:

- **The Elite:** High scores, Project-Based, often Strict.
- **The Popular:** High scores, Easy Grading, Low Workload.
- **The Risk Group:** Low scores, Unclear Grading policies, Poor Communication.

7.2 Challenges & Limitations

1. Domain Shift (The "Digikala Effect")

- **Issue:** The ParsBERT model was pre-trained on Digikala (e-commerce) reviews.
- **Impact:** Initially, the model struggled with educational nuances (e.g., interpreting "Nothing to add" as Negative instead of Neutral).

- **Mitigation:** We fine-tuned the model on our specific dataset, but a larger educational corpus would further improve "Neutral" detection accuracy.

2. Data Imbalance

- **Issue:** As shown in the Initial Audit (Figure 2.1), student ratings are positively skewed (Mean: 7.64/10).
- **Impact:** The model has fewer examples of "Negative" writing styles compared to "Positive" ones, potentially making it slightly less sensitive to subtle complaints.

7.3 Final Verdict & Future Work

Elmos Yar has successfully bridged the gap between raw text and actionable decision-making.

- **For Students:** The **Hybrid Recommender** (Notebook 08) eliminates the anxiety of "blind" course selection by offering personalized matches based on learning style (e.g., "Project-focused" vs. "Exam-focused").
- **For Administration:** The **Clustering Analysis** (Notebook 07) provides an automated, unbiased method to identify underperforming instructors or potential "Grade Inflation" anomalies.

Future Directions (Phase 4):

1. **Deployment:** Hosting the Streamlit dashboard on a university server for public access.
2. **RAG Integration:** Implementing "Retrieval-Augmented Generation" to allow students to chat with the data (e.g., "Summarize the complaints about Dr. X").
3. **Real-time Pipeline:** Automating the data ingestion to update results every semester automatically.